Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: TP05-Part2

Subject: **Operating System**

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For security reasons, the following modifications will need to be made to each of the departments' respective directorie:

- Ensure that the owner of each of the directories is the department administrator and the group ownership is the group for each department

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chown adminsale:Sale Sale
```

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chown adminIS:IS IS
```

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chown admineng Engineering
```

- The department administrator will have full access to their respective department directories
- Ensure that only the owner of a file in the department's directory can delete the file (Sticky bit) . The user will also have ownership of their respective department folders.

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod +t Engineering/
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ ls -l | grep Engineering
drwxrwx--t  2      770 Engineering        4096 Dec 13 14:03 Engineering
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chown adminsale:Sale Sale
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod 770 Engineering
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod 770 Sale
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod +t Sale
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chown adminIS:IS IS
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod 770 IS
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod +t IS
```

- Normal users in each department will have full access (Read, Write and Execute) to their respective department folders.

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo ls -l
total 267192
-rw-rw-r--  1 sokmeak   sokmeak      124238934 Nov 18 01:07 dbeaver-ce_latest_amd64.deb
drwxr-xr-x 13 sokmeak   sokmeak           4096 Dec  6 16:10 Desktop
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov 29 14:06 Documents
drwxr-xr-x  3 sokmeak   sokmeak           4096 Dec 13 14:23 Downloads
drwxrws---  2 admineng  Engineering       4096 Dec 13 14:39 Engineering
drwxrws--T  2 adminIS   IS                4096 Dec 13 14:46 IS
drwxrwxr-x  3 sokmeak   sokmeak           4096 Dec 13 13:46 Lab05
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov  1 10:04 Music
-rwxr-xr--  1 rodri     i4students-c         0 Dec 13 13:15 my_cool_file
-rw-rw-r--  1 sokmeak   sokmeak              0 Dec 13 13:12 my_file.txt
drwxr-xr-x  2 sokmeak   sokmeak           4096 Dec  3 10:03 Pictures
drwxrwxr-x  3 sokmeak   sokmeak           4096 Nov 20 14:31 Postman
-rw-rw-r--  1 sokmeak   sokmeak      149293122 Nov  8 13:37 Postman-linux-x64.tar.gz
drwxrwxr-x  3 sokmeak   sokmeak           4096 Nov 27 12:18 projects
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov  1 10:04 Public
drwxrws--T  2 adminsale Sale              4096 Dec 13 14:45 Sale
drwxr-xr-x  8 root      root              4096 Nov 29 14:26 share
drwx------  3 sokmeak   sokmeak           4096 Nov 27 12:02 snap
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov  1 10:04 Templates
-rw-rw-r--  1 sokmeak   sokmeak             33 Nov 13 15:41 test1.txt
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov  1 10:04 Videos
```

- The department folders will ONLY be accessible by users/administrators in each of the respective departments. Ensure that no one else will have permissions to the folders.
    - Verify:

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ cd Sale
bash: cd: Sale: Permission denied
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ cd Engineering
bash: cd: Engineering: Permission denied
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ cd IS
bash: cd: IS: Permission denied
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$
```

Create a document in each of the department directories.

- The ownerships on this file will be the same as the directory it is located in (SGID).

```
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod g+s IS
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod g+s Engineering
sokmeak@sokmeak-IdeaPad-3-14ITL05:~$ sudo chmod g+s Sale
```

```
drwxrws---  2 admineng  Engineering       4096 Dec 13 14:39 Engineering
-rw-r--r--  1 eng1      sokmeak              0 Dec 14 07:04 file_eng
drwxrws---  2 adminIS   IS                4096 Dec 13 14:46 IS
drwxrwxr-x  3 sokmeak   sokmeak           4096 Dec 13 13:46 Lab05
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov  1 10:04 Music
-rwxr-xr--  1 rodri     i4students-c         0 Dec 13 13:15 my_cool_file
-rw-rw-r--  1 sokmeak   sokmeak              0 Dec 13 13:12 my_file.txt
drwxr-xr-x  2 sokmeak   sokmeak           4096 Dec  3 10:03 Pictures
drwxrwxr-x  3 sokmeak   sokmeak           4096 Nov 20 14:31 Postman
-rw-rw-r--  1 sokmeak   sokmeak      149293122 Nov  8 13:37 Postman-linux-x64.tar.gz
drwxrwxr-x  3 sokmeak   sokmeak           4096 Nov 27 12:18 projects
drwxr-xr-x  2 sokmeak   sokmeak           4096 Nov  1 10:04 Public
drwxrws---  2 adminsale Sale              4096 Dec 13 14:45 Sale
```

- The document should contain only one line of text that states, “This file contains confidential information for the department.”

```
admineng@sokmeak-IdeaPad-3-14ITL05:~$ touch DocEngineering
admineng@sokmeak-IdeaPad-3-14ITL05:~$ echo "This document contains the information of the Engineering Department" > DocEngineering
admineng@sokmeak-IdeaPad-3-14ITL05:~$ cat DocEngineering
This document contains the information of the Engineering Department
```

```
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ echo "This file contains the information of the Sale Department." > DocSale
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ ls
DocSale
```

```
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ echo "This file contains the information of IS department" > DocIS
```

- This file can be read by any user in the department, but can only be modified by the department administrator. No one else has permissions to this file.

```
admineng@sokmeak-IdeaPad-3-14ITL05:~$ touch DocEngineering
admineng@sokmeak-IdeaPad-3-14ITL05:~$ echo "This document contains the information of the Engineering Department" > DocEngineer
ing
admineng@sokmeak-IdeaPad-3-14ITL05:~$ cat DocEngineering
This document contains the information of the Engineering Department
admineng@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rw-r--r-- 1 admineng Engineering 69 Dec 14 06:35 DocEngineering
admineng@sokmeak-IdeaPad-3-14ITL05:~$ chmod 740 DocEngineerig
chmod: cannot access 'DocEngineerig': No such file or directory
admineng@sokmeak-IdeaPad-3-14ITL05:~$ chmod 740 DocEngineering
admineng@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rwxr----- 1 admineng Engineering 69 Dec 14 06:35 DocEngineering
admineng@sokmeak-IdeaPad-3-14ITL05:~$ chmod t DocEngineering
chmod: invalid mode: ‘t’
Try 'chmod --help' for more information.
admineng@sokmeak-IdeaPad-3-14ITL05:~$ chmod +t DocEngineering

admineng@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rwxr----T 1 admineng Engineering 69 Dec 14 06:35 DocEngineering
```

```
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ echo "This file contains the information of the Sale Department." > DocSale
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ ls
DocSale
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rw-r--r-- 1 adminsale Sale 59 Dec 14 06:48 DocSale
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ chmod 740 DocSale
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ chmod +t DocSale

adminsale@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rwxr----T 1 adminsale Sale 59 Dec 14 06:48 DocSale
adminsale@sokmeak-IdeaPad-3-14ITL05:~$ exit
```

```
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ echo "This file contains the information of IS department" > DocIS
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ ls -o
total 4
-rw-r--r-- 1 adminIS 52 Dec 14 06:50 DocIS
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rw-r--r-- 1 adminIS IS 52 Dec 14 06:50 DocIS
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ chmod 740 DocIS
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rwxr----- 1 adminIS IS 52 Dec 14 06:50 DocIS
adminIS@sokmeak-IdeaPad-3-14ITL05:~$ chmod +t DocIS

adminIS@sokmeak-IdeaPad-3-14ITL05:~$ ls -l
total 4
-rwxr----T 1 adminIS IS 52 Dec 14 06:50 DocIS
```